4. IP, MAC, DNS, DHCP and Ports

Basic definitions

Term	What it is	Simple example
IP address	Logical address of a device on a network. Can change. IPv4 is 32-bit, IPv6 is 128-bit.	192.168.1.10
MAC address	Physical hardware address of the network card. 48-bit, set by the manufacturer, works only within the local network.	00:1A:2B:3C:4D:5E
DNS	Converts domain names into IP addresses so people don't have to remember numbers.	google.com becomes 142.250.x.x
DHCP	Automatically gives IP address, gateway and DNS to devices when they join a network.	Your phone gets an IP when it connects to Wi-Fi
Port	A number that identifies a specific service on a device. Range is 0 to 65535.	Port 443 for HTTPS

Points worth remembering

- IP is like your home address, MAC is like your name on your Aadhaar. The address can change, the name stays.
- DHCP works in four steps: Discover, Offer, Request, Acknowledge (DORA).
- DNS uses port 53, mostly UDP, and TCP for large responses and zone transfers.
- An attacker on the same network can fake DHCP or DNS replies (DHCP spoofing, DNS poisoning).

12 common ports

Port	Service	Typical risk
21	FTP	Sends credentials in plain text, anonymous login often left on
22	SSH	Brute force attacks, weak passwords
23	Telnet	No encryption at all, easy to sniff, should be disabled
25	SMTP	Spam relay, email spoofing, phishing
53	DNS	DNS poisoning, tunneling, amplification DDoS
80	HTTP	Unencrypted traffic, web attacks like SQLi and XSS
110	POP3	Plain text mail login if not using TLS
143	IMAP	Plain text mail login if not using TLS
443	HTTPS	Weak TLS configuration, web app vulnerabilities
445	SMB	Very risky if exposed. EternalBlue and

Port	Service	Typical risk
		WannaCry used this port
3306	MySQL	Exposed database, weak or default root password
3389	RDP	Brute force and ransomware entry point, BlueKeep